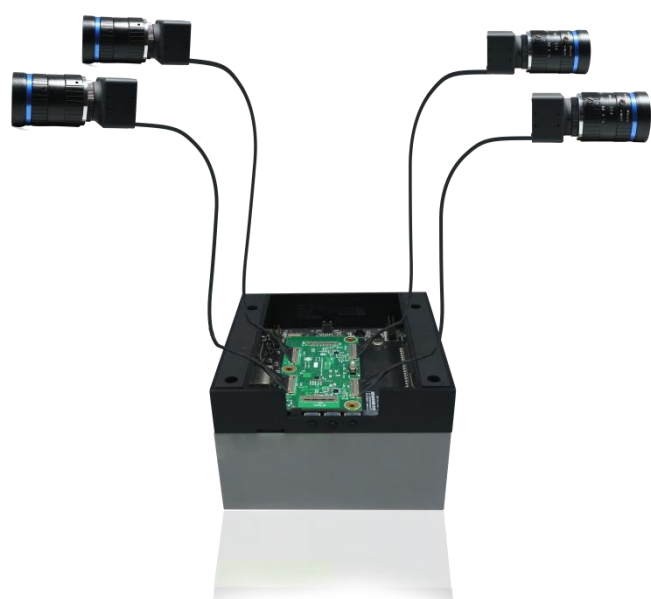


e-CAM82 CUOAGX

Linux Application User Manual



Version 1.4

e-con Systems

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Disclaimer

e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

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Introduction to e-CAM82_CUOAGX

e-CAM82_CUOAGX is an 8 MP customer lens camera module designed and developed by e-con Systems, a leading Embedded Product Design Services Company which specializes in advanced camera solutions. e-CAM82_CUOAGX targets the NVIDIA® Jetson AGX Orin™/ Xavier™ development kits.

e-con Systems provides a sample application, called eCAM_argus_camera, a customized version of argus_camera sample application provided by NVIDIA®, along with the e-CAM82_CUOAGX camera. It is a video viewer software customized to demonstrate the features of e-CAM82_CUOAGX. The main feature of this application is, all connected cameras can be simultaneously streamed and recorded in synchronous (maximum four cameras) and asynchronous mode.

This document describes the usage of eCAM_argus_camera application, and the special features of eCAM_argus_camera application when it is used with e-CAM82_CUOAGX.

Description

Each of the IMX485 based camera in e-CAM82_CUOAGX can stream two resolution, bit depth and frame rates in 4-lane mode in Jetson AGX Xavier™/ Orin™. Each camera can be independently configured to one of the configurations as listed in following tables.

Table 1: Supported Resolutions and Frame Rates in 4-Lane Mode and 2-Lane Mode (Jetson AGX Orin/Xavier)

Lanes	Resolution	Frame Rate(fps) in 10-bit Output	Frame Rate(fps) in 12-bit Output
4-lane	1920 x 1080	90	90
	3840 x 2160	60	50
2-lane	1920 x 1080	60	60
	3840 x 2160	39	33

The camera controls of e-CAM82_CUOAGX are as follows:

- Exposure Time Range
- Gain Range
- ISP Digital Gain Range
- Sensor Mode Index
- Frame Rate
- De-Noise Mode
- De-Noise Strength
- Edge Enhance Mode
- Edge Enhance Strength

- Auto Exposure (AE) Antibanding Mode
- Auto White Balance (AWB) Mode
- Exposure Compensation
- AE Lock
- AWB Lock
- Output Size

The eCAM_argus_camera is a sample libArgus based camera application used to demonstrate the functioning of e-CAM84_CUMI485 module boards in Jetson AGX Xavier™ / Jetson AGX Orin™ boards.

Using eCAM_argus_camera application, you can perform the following:

- Enumerate and list all video capture devices connected.
- Stream all available resolutions, if different resolutions are supported by the device.
- Change controls for all available cameras.
- Capture images in RAW and JPEG formats.
- Record H264, H265 encoded videos on Jetson AGX Xavier™ / Jetson AGX Orin™ boards.

e-con Systems provides prebuilt binary of the application along with source code. Please refer to the *e-CAM82_CUOAGX_Release_Notes.pdf* for the compatible Linux distribution version (L4T version).

Launching the Application

This section describes how to launch the eCAM_argus_camera application.

The commands and output messages in this manual are represented by different colors as listed in following table.

Table 2: Notation of Colors

Color	Notation
Blue	Commands running in development board
Red	Output message in development board

The steps to launch the application are as follows:

1. Connect the e-CAM82_CUOAGX Camera to the J509 connector of the Jetson AGX Xavier™/ Orin™ development kit.
2. Run the following command to check whether all the cameras are initialized.

```
$ sudo dmesg | grep "Detected eimx485 sensor"
```

The output message appears as shown below.

```
Detected eimx485 sensor
```

The output messages indicate that all the cameras are initialized properly.

3. Run the following command to check the presence of video nodes.

```
$ ls /dev/video*
```

The output message for four camera setup appears as shown below.

```
/dev/video0
/dev/video1
/dev/video2
/dev/video3
```

Note: The number of video node entries and number of connected cameras must be same.

4. Run the following commands to achieve maximum frame rate before launching the eCAM_argus_camera application in the Jetson™ board.

```
$ sudo jetson_clocks
$ sudo /home/max-isp-vi-clks.sh
$ sudo nvpmode1 -m 0
```

5. Run the following command to launch the eCAM_argus_camera application.

```
$ eCAM_argus_camera --device=<n>
```

Where <n> refers to the device node.

When entering the command in terminal, the screen appears as shown below.

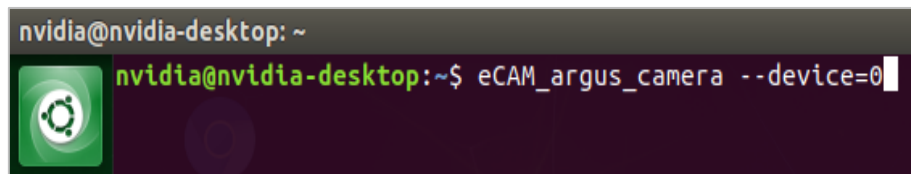


Figure 1: Terminal Window

Once the application is launched, the control window and video display window appear as shown below.

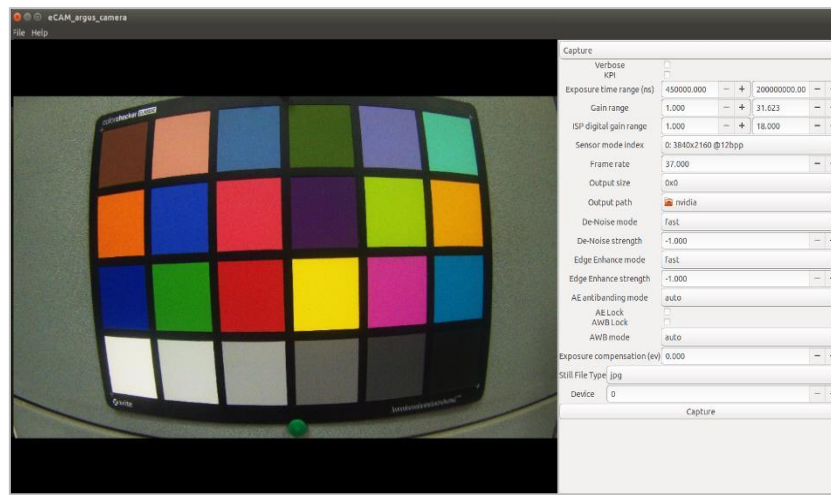


Figure 2: Initial Control Window and Video Display Window Appearance

Application Features

This section describes the controls that are supported in the current version of eCAM_argus_camera application for e-CAM82_CUOAGX.

In eCAM_argus_camera application, the sensor and camera ISP features appear as shown below.

Capture					
Verbose KPI	☐				
Exposure time range (ns)	450000.000	–	+	400000000.00	– +
Gain range	1.000	–	+	31.623	– +
ISP digital gain range	1.000	–	+	256.000	– +
Sensor mode index	0: 3840x2160 @12bpp				
Frame rate	60.000	– +			
Output size	0x0				▼
Output path	nvidia				▼
De-Noise mode	fast				▼
De-Noise strength	-1.000	– +			
Edge Enhance mode	fast				▼
Edge Enhance strength	-1.000	– +			
AE antibanding mode	auto				▼
AE Lock	☐				
AWB Lock	☐				
AWB mode	auto				▼
Exposure compensation (ev)	0.000	– +			
Still File Type	jpg				▼
Device	0	– +			
Capture					

Figure 3: Controls of e-CAM82_CUOAGX

You can click the respective control box and configure the preview settings according to your needs. The value being set will be displayed in the respective text

box. Once the value is set to the respective control box, you can notice the change in preview's property.

The values of e-CAM82_CUOAGX controls are listed in below table.

Table 3: Values of e-CAM82_CUOAGX Controls

Controls	Minimum Value	Maximum Value	Default Value	Manual Control	Auto Control
Exposure time range (ns)	450000	400000000	Auto	Yes	Yes
Gain range (A_f)	1	31.623	-	Yes	Yes
ISP digital gain range	1	256	-	Yes	Yes
Sensor mode index	0	3	0	Yes	No
Frame rate (fps)	2.5	based on sensor mode(refer Table 1)	based on sensor mode((refer Table 1)	Yes	No
De-Noise mode	Off	High quality	Fast	Yes	No
De-Noise strength	-1.00	1.00	-1.00	Yes	No
Edge enhance mode	Off	High quality	Fast	Yes	No
Edge enhance strength	-1.00	1.00	-1.00	Yes	No
AE antibanding mode	Off	60 Hz	Auto	Yes	Yes
AWB mode	Off	Manual	Auto	Yes	Yes
Exposure compensation (eV)	-2	+2	0	Yes	No

Note: The application currently does not support focus position control since the camera has a custom lens which uses only manual focus.

Sensor Features

The sensor features that are supported in the current version of eCAM_argus_camera application are as follows:

- [Exposure Time Range](#)
- [Gain Range](#)
- [Sensor Mode Index](#)
- [Frame Rate](#)

Exposure Time Range

The exposure time range can be changed from a minimum value of 450000 to a maximum value of 400000000 by manually inserting the values or selecting the value in **Exposure time range (ns)** box. The exposure time is represented in Nanoseconds (ns) scale. The default exposure time range set by the application depends upon the luminance of the source. However, to set a specific exposure time

range value, you can manually set both the lower and higher range fields to that same specific value in **Exposure time range (ns)** box.

You can view the screen similar to the screen shown below.

Verbose KPI	☐					
Exposure time range (ns)	450000.000	–	+	4000000000.00	–	+
Gain range	1.000	–	+	31.623	–	+
ISP digital gain range	1.000	–	+	256.000	–	+

Figure 4: Exposure Time Range

Note: The exposure control is limited to the frame rate as the application is designed to achieve a constant frame rate. For example, the maximum exposure that can be achieved is limited to maximum of 40 ms even a larger exposure range value is specified in the **Exposure time range (ns)** box, when the frame rate control is set to 25 fps.

Gain Range

The gain range which corresponds to sensor gain range can be changed from a minimum value of 1 to a maximum value of 31.623 is represented in amplification factor (A_f). IMX485 can support a sensor gain range of 1 to 72 dB corresponding to 0 to 3981.0717 (A_f). However, in e-CAM82_CUOAGX, sensor gain range has been restricted to sensor analog gain, that is. 0 - 30 dB, including the image quality and other performance metrics. The default gain value set depends upon the luminance of the source. However, to set a specific constant sensor gain value, you can manually set both the lower and higher range fields to that same specific value in **Gain range** box. You can view the screen similar to the screen shown below.

Exposure time range (ns)	450000.000	–	+	4000000000.00	–	+
Gain range	1.000	–	+	31.623	–	+
ISP digital gain range	1.000	–	+	256.000	–	+
Sensor mode index	0: 3840x2160 @12bpp ▼					

Figure 5: Gain Range in Jetson AGX Xavier/Orin

Note: The sensor gain range and exposure time range changes are local. On changing the modes, the corresponding values will be discarded, and the default range values will be applied.

Sensor Mode Index

The sensor mode index contains all the combinations of resolutions, frame rate and bpp (bits per pixel) that the camera sensor outputs. e-CAM82_CUOAGX currently

supports the following sensor modes in 4-lane setup (in Jetson AGX Xavier™ and Jetson AGX Orin™) which is listed in below tables.

Table 4: Modes and Resolutions supported in 4-Lane and 2-Lane in Jetson AGX Orin/Xavier

Lanes	Resolution	Frame Rate(fps) in 10-bit Output	Frame Rate(fps) in 12-bit Output
4-lane	1920 x 1080	90	90
	3840 x 2160	60	50
2-lane	1920 x 1080	62	62
	3840 x 2160	39	33

Note: The output from the NVIDIA® ISP is NV12 format.

You can change the sensor modes from the **Sensor mode index** drop-down list box. When you click the **Sensor mode index** drop-down list box, you can view the list of sensor modes as shown below.



Figure 6: Sensor Mode Index

Frame Rate

The frame rate control can be changed from a minimum value of 2.5 to the maximum value specified by the sensor mode (Refer Table). The frame rate can be changed by manually inserting the value or selecting the value in **Frame rate** box. You can view the screen similar to the screen shown below.



Figure 7: Frame Rate

Note: Frame rate displayed (when KPI is enabled) is the number of frames from all streaming cameras per second. For example, 3840 x 2160 at 60 fps mode, when two cameras are streaming, the FPS displayed will be $(30 \times 2) = 60$ fps. Also, the FPS displayed will not go beyond the refresh rate of the monitor used.

Camera ISP Features

The camera ISP features supported in the current release version of eCAM_argus_camera application are as follows:

- [ISP Digital Gain Range](#)
- [De-Noise Mode](#)
- [De-Noise Strength](#)
- [Edge Enhance Mode](#)
- [Edge Enhance Strength](#)
- [AE Antibanding Mode](#)
- [AWB Mode](#)
- [Exposure Compensation](#)
- [Output Size](#)
- [AWB Lock](#)
- [AE Lock](#)
- [Still File Type](#)
- [Video Recording](#)
- [Multi Session](#)
- [Multi Exposure](#)

ISP Digital Gain Range

The ISP gain control can be changed from a minimum value of 1 to a maximum of 18. This gain corresponds to the digital gain additionally provided by the ISP (over and above the sensor gain) and the default value depends upon the luminance of the source. Jetson AGX Xavier™ /Orin™ ISP can support a digital gain range of 1 to 256 (A_f) corresponding to 0 to 48.16 dB. However, in e-CAM82_CUOAGX, ISP gain has been restricted to 1 - 256 (A_f), including the image quality and other performance metrics. To manually set the ISP digital gain, you can change the lower range value and the higher range value to a constant in the **ISP digital gain range** box. You can view the screen similar to the screen shown below.

Exposure time range (ns)	450000.000	–	+	4000000000.00	–	+
Gain range	1.000	–	+	31.623	–	+
ISP digital gain range	1.000	–	+	256.000	–	+
Sensor mode index	0: 3840x2160 @12bpp ▼					

Figure 8: ISP Digital Gain Range

De-Noise Mode

The eCAM_argus_camera application supports the following de-noise modes:

- Off
- Fast

- High Quality

You can select the required de-noise mode from the **De-Noise mode** drop-down list box. The default de-noise mode is fast. When you click the **De-Noise mode** drop-down list box, the supported de-noise modes will be listed as shown below.

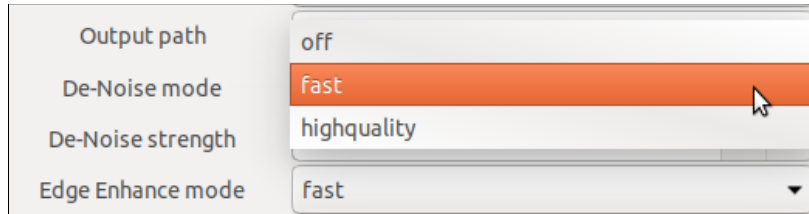


Figure 9: De-Noise Mode

De-Noise Strength

You can change the de-noise strength value from a minimum value of -1.0 to a maximum value of 1.0. The default de-noise strength value is -1.0 and can be changed by manually inserting the values or selecting the values in **De-Noise strength** box. You can view the screen similar to the screen shown below.

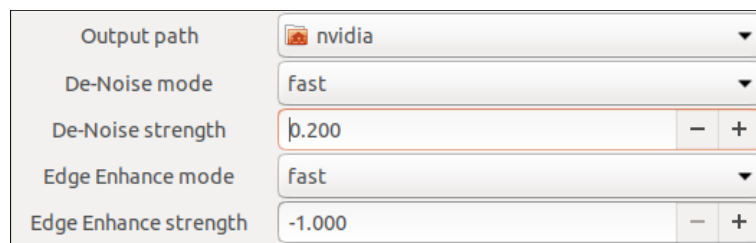


Figure 10: De-Noise Strength

Edge Enhance Mode

The eCAM_argus_camera application supports the following edge enhance modes:

- Off
- Fast
- High Quality

You can select the required edge enhancement mode from the **Edge Enhance mode** drop-down list box. The default edge-enhance mode is fast. When you click the **Edge Enhance mode** drop-down list box, the supported edge enhancement modes will be listed as shown below.

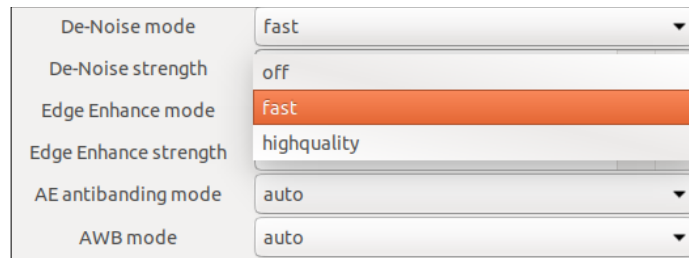


Figure 11: Edge Enhance Mode

Edge Enhance Strength

You can change the edge enhance strength value from a minimum value of -1.0 to a maximum value of 1.0. The default edge enhance strength value is -1.0 and can be changed by manually inserting the values or selecting the values in **Edge Enhance strength** box. You can view the screen similar to the screen shown below.



Figure 12: Edge Enhance Strength

AE Antibanding Mode

The eCAM_argus_camera application supports the AE antibanding modes as follows:

- Off
- Auto
- 50 Hz
- 60 Hz

By default, the AE antibanding mode is auto. You can select the required AE antibanding mode from the **AE antibanding mode** drop-down list box. When you click the **AE antibanding mode** drop-down list box, the list of AE antibanding modes appear as shown below.

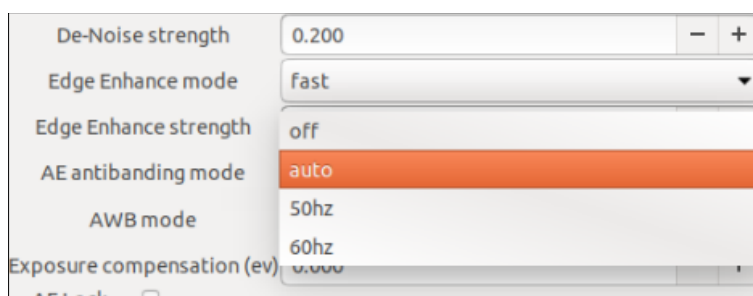


Figure 13: AE Antibanding Mode

AWB Mode

The eCAM_argus_camera application automatically detects the scene temperature and adjusts the white balance. The application supports the following AWB modes:

- Off
- Auto
- Incandescent
- Fluorescent
- Warm Fluorescent
- Daylight
- Cloudy Daylight
- Twilight
- Shade
- Manual

By default, auto mode is enabled. You can select the required AWB modes from the **AWB mode** drop-down list box.

When you click the **AWB mode** drop-down list box, the list of AWB modes appear as shown below.

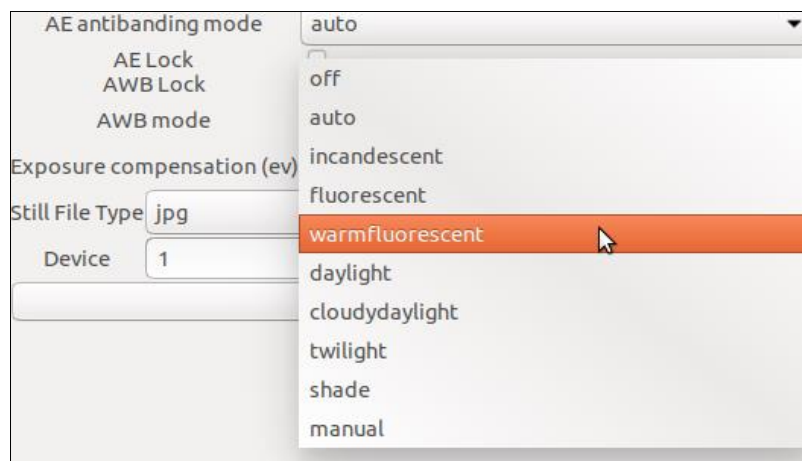


Figure 14: AWB Mode

Exposure Compensation

The exposure compensation control is used to set the threshold of the optimum exposed image. This control value can be changed from a minimum value of -2.00 ev to a maximum value of +2.00 ev. The default value for exposure compensation is 0.00 ev.

You can change the exposure compensation value by manually inserting the values or selecting the value from the **Exposure compensation (ev)** box as shown below.

AWB mode	auto	
Exposure compensation (ev)	0.400	- +
Still File Type	jpg	
Device	1	- +

Figure 15: Exposure Compensation

Output Size

When you click the **Output size** drop-down list box, you can view the supported output sizes as shown below.

Frame rate	37.000	- +
Output size	0x0	
Output path	176x144	
De-Noise mode	320x240	
De-Noise strength	640x480	
Edge Enhance mode	1280x720	
Edge Enhance strength	1920x1080	
	3840x2160	

Figure 16: Output Size

Note: 0x0 Output size denotes the original size

AE Lock

The AE lock control is enabled during the capture and video module of eCAM_argus_camera. When you select the **AE Lock** check box, it locks the exposure to current values. By default, the AE lock is set when the application runs in multi session mode. The AE Lock will appear as shown below.

AE Lock	☐
---------	--------------------------

Figure 17: AE Lock

AWB Lock

When you select the **AWB Lock** check box, it locks the AWB to current value. The AWB Lock option will appear as shown below.

AWB Lock	☒
----------	-------------------------------------

Figure 18: AWB Lock

Still File Type

There are two types of still capture supported in eCAM_argus_camera. You can select the type of still image capture to be JPEG compressed or uncompressed ISP processed YUV frame using the **Still File Type** drop-down list box provided in the Capture module of eCAM_argus_camera.

When you click the **Still File Type** drop-down list box, you can view the supported still file type as shown below.

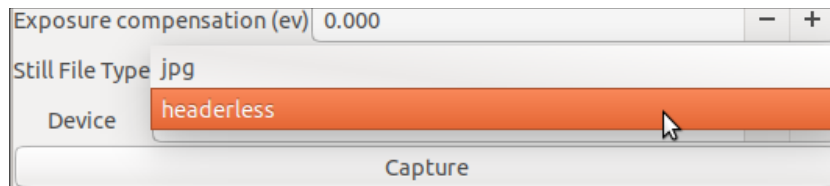


Figure 19: Still File Type

Video Recording

The eCAM_argus_camera also provides the feature for recording video.

When you click the **Video Format** drop-down list box, the list of supported formats for Jetson AGX Orin™ and Jetson AGX Xavier™ will appear as shown in below figures.



Figure 20: Video Recording in Jetson Xavier and Orin

The supported video formats supported by eCAM_argus_camera are as follows:

- H264
- H265

When you click the **Video File Type** drop-down list box, the list of supported file types will appear as shown below.

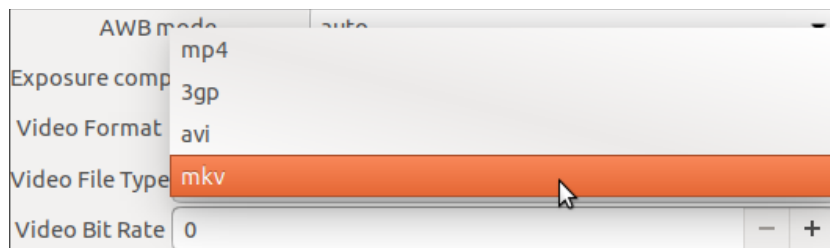


Figure 21: Video File Type

The supported video file types are as follows:

- MP4
- 3GP
- AVI
- MKV

All the above-mentioned video file types support H264 video encoding format only. Only **mkv** video file type supports **vp9** and **h265** video encoding formats. Only **3GP** doesn't support **vp8** format.

Note: To view the recorded videos in Jetson AGX Xavier™ /Jetson AGX Orin™, please ensure the gstreamer plugins are properly installed. Please refer to the e-CAM82_CUOAGX_*Gstreamer_Usage_Guide.pdf* to download the required plugins.

Multi Session

The customized eCAM_argus_camera application provides multi session feature to run multiple cameras simultaneously. You can enable the multi session mode from the drop-down list box provided in the GTK.

When you click the **Multi Session** drop-down list box provided in the GTK, the available modes will appear as shown below.

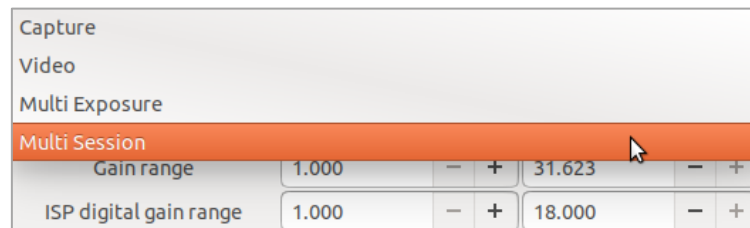


Figure 22: Multi Session Modes

Multi Exposure

The customized eCAM_argus_camera application provides multi exposure feature to provide frames with various exposure simultaneously. You can enable the multi exposure mode from the **Multi Exposure** drop-down list box provided in the GTK.

When you click the **Multi Exposure** drop-down list box provided in the GTK, the available modes will appear as shown below.

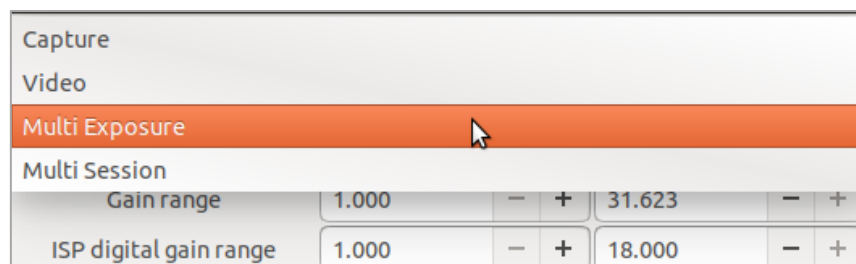


Figure 23: Multi Exposure Mode

For instance, when selected multi exposure mode, it would display three windows with frames from same camera at different exposure. The eCAM_argus_camera application will be displayed as shown below.

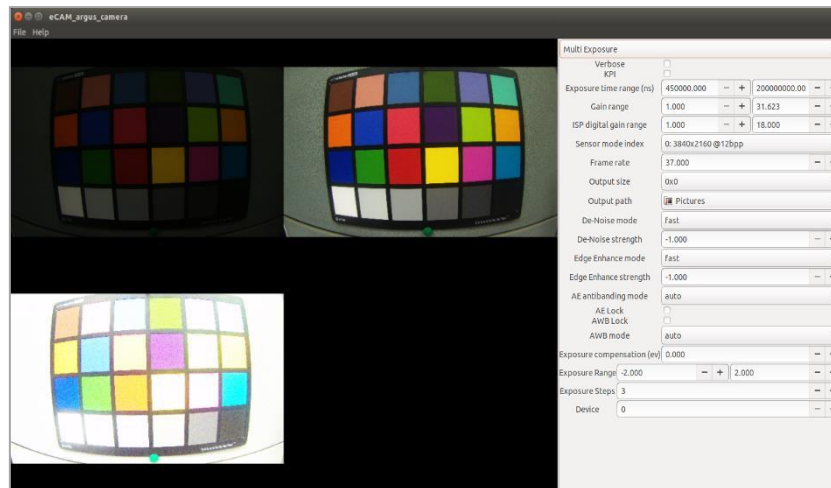


Figure 24: Window Appearance in Multi Exposure Mode

Note: When the application run in multi exposure mode, the AE Lock is set in the background due to limitation in NVIDIA® ISP. Also, FPS is low when capturing multiple frames at different exposure.

Exposure range can be set from -10 to 10 range in which -10 is least exposure and 10 is highest exposure. Exposure steps can be set to maximum of six frames which denotes the number of frame windows to capture frames at different exposures.

Comparing eCAM_argus_camera with argus_camera by NVIDIA

This section describes the comparison of eCAM_argus_camera with argus_camera.

Users familiar with NVIDIA® Jetson AGX Xavier™ /Jetson AGX Orin™ board may be familiar with NVIDIA® default camera application, argus_camera.

Additional features in NVIDIA® argus_camera and not in e-con Systems eCAM_argus_camera are as follows:

- YUV Format
- Focus Position
- Aperture motor step
- Aperture motor speed

Troubleshooting

In this section, you can view the list of commonly occurring issues and their troubleshooting steps.

Recording the video closes the sample application

This is a known limitation. Recording the video in any supported resolution more than once will close the application.

To recover the application, run the following command.

```
$ sudo service nvargus-daemon restart
```

When four cameras connected multi session mode doesn't work properly.

This is a known limitation. This limitation is because of Argus application provided with Jetpack5.1.

In any supported resolutions captured images are corrupted when output size is not 0x0

This is a known limitation. This limitation is because of Argus application provided with Jetpack5.1.

v4l2-compliance test fails in buffer ioctl

VIDIOC_REQBUFS ioctl fail in v4l2-compliance test. This fail is because of Nvidia's V4L2 framework implementation.

Streams stops randomly when running the cameras for longer days.

This is a known issue.

To recover the stream, run the following command and re-launch the application

```
$ sudo service nvargus-daemon restart
```

1. Why auto mode in AE antibanding is not working?

This is a known issue in NVIDIA® ISP. For any other support, please write to camerasolutions@e-consystems.com with your requirement.

What's Next?

After understanding the usage of eCAM_argus_camera application, you can refer to the following documents to understand more about e-CAM82_CUOAGX.

- *e-CAM82_CUOAGX-eCAM_Argus_Camera_Build_and_Install_Guide*
- *e-CAM82_CUOAGX Datasheet*
- *e-CAM82_CUOAGX Release Notes*

Glossary

GIMP: GNU Image Manipulation Program.

GNU: GNU's Not Unix.

GTK: GIMP Toolkit.

ISP: Image Signal Processor.

MIPI: Mobile Industry Processor Interface.

Support

Contact Us

If you need any support on e-CAM82_CUOAGX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	09-Nov-2022	Initial Draft	Camera Dev Team
1.1	16-Nov-2022	Added support for Xavier	Camera Dev Team
1.2	22-Feb-2023	4-lane support	Camera Dev Team
1.3	10-Mar-2023	Updated Application launching section	Camera Dev Team
1.4	14-Mar-2023	Updated L4T35.2.1 version	Camera Dev Team